

Design FMEA - Smartphone Battery

Case ID: VBF-T6R1MIMO-DFMEA-001

Date: 2026-08-31

Status: Design Not Achieved

1. FMEA Summary

Item	Failure Mode	Effect	Cause	Current Controls	RPN	Recommended Actions
1	Lithium plating during 4C charge	Internal short, capacity fade, safety hazard	Low electrol			
2	Excessive temperature during 4C charge	Thermal degradation, separator shutdown, capacity fad				
3	Low energy density	Reduced runtime, competitive disadvantage	Chen2020 system ceiling, elec			
4	Low voltage plateau	Reduced energy, power limitations	NMC811 chemistry limits	None	4	
5	SEI growth exceeding limits	Capacity fade, impedance rise	Electrolyte decomposition, cycli			
6	Electrolyte leakage	Cell failure, safety hazard	Seal failure, manufacturing defect	None		
7	Electrode delamination	Capacity fade, impedance rise	Mechanical stress, adhesion failure			

2. Detailed Failure Mode Analysis

Item 1: Lithium Plating During 4C Charge

****Failure Mode**:** Lithium metal deposits on anode surface during fast charge

****Effect**:**

- Internal short circuit risk
- Capacity fade (lithium inventory loss)
- Safety hazard (thermal runaway potential)

****Cause**:**

- Low electrolyte transport (diffusivity, conductivity)
- High anode overpotential
- Insufficient Li⁺ diffusion in electrode

****Current Controls**:**

- High-conductivity electrolyte ($\sigma=1.8$ S/m, $D=6e-10$ m²/s, $t+=0.45$)
- Anode potential monitoring during charge
- Plating detection algorithm

****RPN Score**:** 8 (Severity=8, Occurrence=1, Detection=1)

****Recommended Actions**:**

- Validate electrolyte conductivity in production
- Implement real-time plating detection in BMS
- Consider electrode microstructure optimization

Item 2: Excessive Temperature During 4C Charge

****Failure Mode**:** Cell temperature exceeds 50°C during 4C fast charge

****Effect**:**

- Thermal degradation of materials
- Separator shutdown (>130°C)
- Capacity fade
- Safety hazard

****Cause**:**

- High current density (4C = 19.8 A)
- Joule heating (I^2R losses)

- Insufficient cooling at 45°C ambient
- 5°C margin to target is physically inadequate

****Current Controls**:**

- Thermal model validation
- Temperature monitoring during charge
- Charge rate limiting based on temperature

****RPN Score**:** 12 (Severity=9, Occurrence=2, Detection=2)

****Recommended Actions**:**

- Relax thermal specification to 80°C for 4C charge
- Implement active cooling system
- Reduce charge rate to 3C when temperature approaches limit
- Validate thermal management system design

Item 3: Low Energy Density

****Failure Mode**:** Volumetric energy density below 950 Wh/L target

****Effect**:**

- Reduced smartphone runtime
- Competitive disadvantage
- Larger battery required for same capacity

****Cause**:**

- Chen2020 system ceiling ~850 Wh/L
- Electrode thickness limited by transport
- Active material density fixed

****Current Controls**:**

- Architecture optimization (porosity, thickness)
- High-conductivity electrolyte for better transport
- Electrode formulation tuning

****RPN Score**:** 6 (Severity=6, Occurrence=2, Detection=1)

****Recommended Actions**:**

- Use OKane2022 parameter set with SiO_x anode
- Explore high-capacity cathode materials
- Optimize electrode microstructure for higher loading

Item 4: Low Voltage Plateau

****Failure Mode**:** Voltage plateau below 4.1V target

****Effect**:**

- Reduced energy density
- Power limitations
- Competitive disadvantage

****Cause**:**

- NMC811 chemistry peaks at ~4.04V
- No high-voltage cathode in current scope
- Operating conditions limit voltage

****Current Controls**:**

- None (chemistry limitation)

****RPN Score**:** 4 (Severity=5, Occurrence=1, Detection=1)

****Recommended Actions**:**

- Use LNMO high-voltage cathode (4.7V class)

- Explore other high-voltage chemistries
- Consider hybrid cathode approaches

Item 5: SEI Growth Exceeding Limits

****Failure Mode**:** SEI thickness exceeds 500 nm after 100 cycles

****Effect**:**

- Capacity fade
- Impedance rise
- Reduced cycle life

****Cause**:**

- Electrolyte decomposition
- Cycling stress
- Temperature effects

****Current Controls**:**

- Baseline SEI model (Chen2020)
- SEI thickness monitoring
- Electrolyte formulation optimization

****RPN Score**:** 2 (Severity=4, Occurrence=1, Detection=1)

****Recommended Actions**:**

- Add FEC film-forming additive
- Validate SEI growth in production
- Monitor capacity fade during cycling

Item 6: Electrolyte Leakage

****Failure Mode**:** Electrolyte leaks from cell

****Effect**:**

- Cell failure
- Safety hazard
- Environmental contamination

****Cause**:**

- Seal failure
- Manufacturing defect
- Mechanical damage

****Current Controls**:**

- None (not modeled in simulation)

****RPN Score**:** 3 (Severity=7, Occurrence=1, Detection=1)

****Recommended Actions**:**

- Add electrolyte leak detection
- Validate seal integrity in production
- Implement mechanical abuse testing

Item 7: Electrode Delamination

****Failure Mode**:** Electrode separates from current collector

****Effect**:**

- Capacity fade
- Impedance rise
- Reduced power capability

****Cause**:**

- Mechanical stress
- Adhesion failure
- Thermal cycling

****Current Controls**:**

- None (not modeled in simulation)

****RPN Score**:** 3 (Severity=5, Occurrence=1, Detection=1)

****Recommended Actions**:**

- Validate adhesion in production
- Implement mechanical abuse testing
- Optimize electrode formulation for adhesion

3. Risk Priority Summary

RPN Range	Count	Items
High (≥ 10)	1	Item 2 (Temperature)
Medium (5-9)	2	Item 1 (Plating), Item 3 (ED)
Low (< 5)	4	Item 4 (Voltage), Item 5 (SEI), Item 6 (Leakage), Item 7 (Delamination)

4. Recommendations

Critical Issues (High RPN)

- **Thermal Management**:** Relax thermal specification or implement active cooling for 4C charge
- **Plating Prevention**:** Validate high-conductivity electrolyte in production

Important Issues (Medium RPN)

- **Energy Density**:** Use high-capacity materials for > 900 Wh/L applications
- **Electrolyte Validation**:** Ensure conductivity meets specification

Minor Issues (Low RPN)

- **Voltage Plateau**:** Consider high-voltage cathode for future designs
- **SEI Control**:** Add FEC additive for improved cycling
- **Manufacturing Validation**:** Implement seal and adhesion testing

5. Sign-off

Role	Name	Date	Signature
Design Engineer	-	2026-08-31	-
Quality Engineer	-	2026-08-31	-
Safety Engineer	-	2026-08-31	-

****Note**:** This is a qualitative DFMEA based on simulation results. Physical validation and production